→ reverse log operation eg. $\ln y = \beta_0 + \beta_1 \ln x_1 + \beta_2 x_2 + u_2$

I. $\Delta x_2 = 1 \Rightarrow \ln y - \ln y_0 = \beta_2 \Rightarrow \frac{y}{y_0} = e^{\beta_2} \Rightarrow \frac{y-y_0}{y_0} = e^{\beta_2} - 1 \approx \beta_2$

II. $\Delta x_1\% = 1 \Rightarrow \frac{x_1}{x_{1,0}} = 1.01 \Rightarrow \ln\frac{y}{y_0} = \beta_1 \ln(1.01) \Rightarrow \frac{y}{y_0} = 1.01^{\beta_1}$

β_1 & β_2 越靠近0越准 $\Rightarrow \frac{y-y_0}{y_0} = 1.01^{\beta_1} - 1 \approx 0.01\beta_1 = \beta_1\%$

→ Predict y when ln(y) $\ln(y) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$.

I. 假设 $u \sim N(0, \sigma^2)$ 则 $E(\exp(u)) = \exp(\frac{\sigma^2}{2})$

∴ $y = \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k) \cdot \exp(u)$

$E(y|x) = \exp(\frac{\sigma^2}{2}) \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)$ $\hat{y} = \exp(\frac{\hat\sigma^2}{2}) \exp(\log \hat{y})$

II. 大样本 $\exp(u) \to \frac{1}{n}\sum \exp(\hat{u}_i)$ $\hat{y} = \frac{\sum \exp(\hat{u}_i)}{n} \exp(\ln \hat{y})$

4. Quadratic & Interaction

① eg $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \beta_{k+1} x_1^2 + u$. $\frac{\partial E(y|x_1)}{\partial x_1} = \beta_1 + 2\beta_{k+1} \cdot x_1$

Δx_1 对 y 的影响取决于 x_1 的值 → 可算 $\beta_1 + 2\beta_{k+1} x_1 = 0$ $x_1 = \dots$ turnaround point

⇒ log x 与 log x² √ perfect collinearity log x & (log x)² ×

⇒ High polynominals 更可能多重共线性问题. 与 x_1 有关

② Interaction $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$ ∴ $\frac{\partial y}{\partial x_2} = \beta_2 + \beta_3 x_1$

5. Function form misspecification test 与 economic theory

① Ramsey Reset Test 在 MLR1 ~ MLR6

则 高阶次 or Interaction 原: $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k \to \hat{y}^2, \hat{y}^3$ stata 看 → 至 $\hat{y}^4$

用 $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \delta_1 \hat{y}^2 + \delta_2 \hat{y}^3$ Joint F-test $\delta_1 = \delta_2 = 0$?

$F \sim F_{2,n-k-3}$ or $LM \sim \chi^2_2$ if reject H_0 则 misspecification

② Nonnested Alternative Test

Model 1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ ⇒ $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 \log(x_1) + \beta_4 \log(x_2) + u$

2: $y = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + u$

⇒ 永远可以做但不一定有结果 eg. β_1 β_2 sig / y & ln y 用不了 因变量需一样

③ DM-Test $\hat{y} = \hat\beta_0 + \hat\beta_1 x_1 + \hat\beta_2 x_2$ $y = \beta_0 + \beta_1 \ln(x_1) + \beta_2 \ln(x_2) + \theta_1 \hat{y} + error$

if θ_1 sig 则 需加入 x_1, x_2 变量

6. Adj R² 原 $R^2 = 1 - \frac{\sum \hat{u}^2}{SST}$, R^2 不代表 validation or casual relation.

加入新 x, R^2 一定至少不变 且是 sample 中 y variation explained by regressor

$R^2 = 1 - \frac{SSR/n}{SST/n}$ estimator for $1 - \frac{\sigma_u^2}{\sigma_y^2}$ 加入新 x ↗ 不变 ... ↘ Error Var ↓

⇒ 应加入 uncorrelated with other regressor 的 x.

Adj $R^2 = 1 - \frac{SSR/(n-k-1)}{SST/(n-1)}$ penalty for 加入 new x. ⇔ Adj R^2 ↑ New x 的 |t| > 1

$\bar{R}^2 = 1 - (1 - R^2)(n-1)/(n-k-1)$ ⇒ non-nested choice 标准.

$R^2 = corr(y_i, \hat{y}_i)^2$ 本前提 y 一样 ⇒ y 变量不同 模型选择

7. Proxy variable for omitted variable bias

① True $y^* = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + u$ reg y^* x_1^* ($y^* = \tilde\alpha x_{i1} + u_i$)

则 $\hat\alpha_1 = \frac{\sum(x_{i1} - \bar{x}_1)(y_i - \bar{y})}{\sum(x_{i1} - \bar{x}_1)^2} = \beta_1 + \beta_2 \frac{\sum(x_{i2} - \bar{x}_2)(x_{i1} - \bar{x}_1)}{\sum(x_{i1} - \bar{x}_1)^2} + \frac{\sum(u_i - \bar{u})(x_{i1} - \bar{x}_1)}{\sum(x_{i1} - \bar{x}_1)^2}$

inconsist & biased (大样本) $\xrightarrow{p} \beta_1 + \beta_2 \frac{cov(x_1, x_2)}{Var(x_1)} + \frac{cov(u, x)}{Var(x_1)}$ = 0

② proxy variable 解决 omitted variable bias 需要 proxy related to omitt

eg. $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3^* + u$ $x_3^* = \delta_0 + \delta_3 x_3 + v_3$.

x_3^* 无法得到 如 ability 用 x_3 IQ 当成 proxy ⇒ plug-in solution

前提 u 与 x_1, x_2, x_3^*, x_3 corr = 0 / v_3 与 x_1, x_2, x_3 corr = 0 /

更易偏 若与 x_1, x_2 无关 与 x_3 有关 $E(x_3^*|x_1, x_2, x_3) = E(x_3^*|x_3) = \delta_0 + \delta_3 x_3$

reg y x_1 x_2 x_3 ⇒ $y = (\beta_0 + \beta_3\delta_0) + \beta_1 x_1 + \beta_2 x_2 + \beta_3\delta_3 x_3 + (u + \beta_3 v_3)$

否则 $x_3^* = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + v$ 代入后 $\beta_1 = \beta_1 + \beta_3\delta_1$ biased

③ 找不到 proxy, 加入 lagged dependent variable y_{t-1} 与 y_t related

8. Measurement Error proxy: 无 measurement 有的还有 error

① Measurement Error in y y^* true / y observed ⇒ $e_0 = y - y^*$

e_0 是 random measurement error

$y = y^* + e_0$ reg $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$ (y^*) $+ e_0$ u'

7. OLS 和 IV 当 $cov(z, u) \neq 0$ 时 IV 也 inconsistent

OLS: $plim\ \hat\beta_1 = \beta_1 + \frac{cov(x, u)}{Var(x)} = \beta_1 + corr(x, u) \cdot \frac{\sigma_u}{\sigma_x}$

IV: $plim\ \hat\beta_1 = \beta_1 + \frac{cov(z, u)}{cov(z, x)} = \beta_1 + \frac{corr(z, u)}{corr(z, x)} \cdot \frac{\sigma_u}{\sigma_x}$

prefer: 看 $\left|\frac{corr(z, u)}{corr(z, x)}\right|$ & corr(x, u) 哪个小 选哪个 (大样本)

8. Multiple regression 中 IV 条件② ⇒ partially correlated w/ endo 变量

eg. $y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \dots + \beta_k z_{k-1} + u_1$ (y_2 endogenous)

$y_2 = \pi_0 + \pi_1 z_1 + \dots + \pi_k z_{k-1} + \pi_k z_k + v_2$ → IV reduced form regression

9. Two stage least square (2SLS Estimation) ⇔ IV

① reduced form regression ⇒ $\hat{y}_2 = \hat\pi_0 + \hat\pi_1 z_1 + \dots + \hat\pi_k z_{k-1} + \hat\pi_k z_k$ IV

② OLS with y_2 replaced by $\hat{y}_2$: $y_1 = \beta_0 + \beta_1 \hat{y}_2 + \beta_2 z_1 + \dots + \beta_k z_{k-1} + error$

∵ $y_2 = \hat{y}_2$ (exo) $+ \hat{v}_2$ (endo) ⇒ error $= u_1 - \beta_1 \hat{v}_2$ 与 $\hat{y}_2$ 无关

⇒ 成功 $\hat{y}_2$ 把 endo 部分去除. / ② 中 所有变量 exo.

10. 1个 endo 1个 IV: 2SLS ⇔ IV   best $y_2 = \pi_0 + \pi_1 z_1 + \dots + \pi_k z_k + \pi_{k+1} z_{k+1} + v_2$

1个 endo 多 IV: 2SLS ① 中 所有 IV 都 放入

多个 endo order condition: IV 量 > endo variable (必要条件)

rank condition: 每个 var 都需有 IV (充分条件)

11. Testing for endogeneous variable.

① $y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \dots + \beta_k z_{k-1} + u_1$ → IV $H_0: \delta_1 = 0$ 时 exo

1° $y_2 = \pi_0 + \pi_1 z_1 + \dots + \pi_k z_{k-1} + \pi_k z_k + v_2$

y_2 exogeneous ⇔ v_2 uncorrelated with u_1 ⇒ $u_1 = \delta_1 v_2 + e_1$

2° 得 $\hat{v}_2$: $y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \dots + \beta_k z_{k-1} + \delta_1 \hat{v}_2 + e_1$

$H_0: \delta_1 = 0$ → exo $H_1: \delta_1 \neq 0$ → endo → 用 IV

② Hausman Test: $H_0: \beta_{OLS} = \beta_{IV}$ 无区别 → exo 用 OLS 更好

H_1 ≠ 显著不同 → endo 用 IV 更好

§9 Time series Analysis

1. Time Series Def: same observational unit at multiple time period

① Temporal ordering × random sampling × arbitrarily reordered

② Typical feature: serial correlation (dependence of obs)

③ randomness 来源: × random sampling from population

√ sequence of random vectors $(V_1, V_2, \dots)$ $V_t = (y_t, x_{t1}, \dots)$ = stochastic process

④ Population: all possible path the stochastic process could have taken

Sample: the one realized path of time series in the population

2. Time series regression model ① static model y_t & x_t

② Finite Distributed lag models (FDL) y_t & $x_t, x_{t-1}, \dots, x_{t-g}$

FDL(2): $y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \dots + \delta_g z_{t-g} + u_t$ FDL(n) n 阶 lag

I. immediate change $\delta_0 = E(y_t) - E(y_{t-1})$ → impact propensity/multiplier reflect immediate change

($z_t = c+1$, 之前 $= c$) $\delta_1 = E(y_{t+1}) - E(y_{t-1})$

$\delta_2 = E(y_{t+2}) - E(y_{t-1})$ one/.../g period lag multiplier

II. permanent shock $\delta_0 = E(y_t) - E(y_{t-1})$; $\delta_1 = E(y_{t+1}) - E(y_{t-1})$

(z_t 之后全是 $c+1$) ...., $\delta_0 + \dots + \delta_g = E(y_{t+s}) - E(y_{t-1})$ for $s \geq g$

Long run propensity: reflect long-run change in y after permanent shock

3. Finite sample properties under Classical Assumptions

TS1: Linear in parameters   MLR2 random sampling

TS2: No perfect collinearity   一定不满足 → 跳过

TS3: Zero conditional mean → exo/endo (过去、现在、未来)

{ strict exogeneity: $E(u_t|X) = 0$ u_t 与 任何 X_t 无关

contemporaneous exo: $E(u_t|X_t) = 0$ only 在 t 时 无关

TS4 Homoskedasticity $Var(u_t|X) = Var(u_t) = \sigma^2$ y 与 regime change violate

TS5 (weaker random sampling) → No serial correlation

$corr(u_t, u_s|X) = 0$ ($t \neq s$) serial correlation 违反

TS4: 对角线 σ^2 ⇒ $Var(u)_{T\times T} = \begin{pmatrix} Var(u_1) & cov(u_1, u_2) \\ cov(u_2, u_1) & Var(u_2) \end{pmatrix} = \begin{pmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{pmatrix}$

TS5: 非对角 0   TS4/5

TS6. Normality $u_t \sim N(0, \sigma^2)$ u_t iid 且 u_t 与 X 独立

8. Asymptotic property of OLS (在 large sample 下)

TS'1 Linear in parameter + stationary & weakly dependent (x & y)

TS'2 No perfect collinearity (含自身 lag 项 一定违反 strict exo)

TS'3 relaxed contemporaneous exogeneity $E(u_t|X_t) = 0$

TS'4 Homoske. $Var(u_t|X_t) = \sigma^2$

TS'5 relaxed No serial correlation $corr(u_t, u_s|X_t, X_s) = 0$

⇒ { TS1'~3' consistency $plim\ \hat\beta_j = \beta_j$ 但是 biased

TS1'~5' OLS asymptotically $\sim N / Se(\hat\beta_j)$, t-test F-test 可用

9. 例外 ① Trend-stationary series nonstationary only cuz time t

eg. $y = \beta_0 + \beta_1 t + u_t$ ⇒ OLS valid weakly-dependent

例外 ② Highly persistent (× weakly dependent √ strong dependent)

⇒ OLS invalid ⇒ transform to weakly dependent: Difference n times I(n)

eg. I(1) $y_t = y_{t-1} + e_t$ (random walk) ⇒ $\Delta y_t = y_t - y_{t-1} = e_t$ weakly-

$\log(y_t) = \log(y_{t-1}) + e_t$ ⇒ $\Delta \log(y_t) = e_t$ dependent

Test for I(1): $\hat\rho_1 = corr(y_t, y_{t-1})$ if $\hat\rho_1$ 接近 1 ⇒ unit root ⇒ Difference

§10 Serial correlation → violate TS5 $corr(u_t, u_s|X) \neq 0$

1. 后果: 仍满足 OLS unbias (TS1~3) / consistent (TS1'~TS3') / R^2

不对: Se, $Var(\hat\beta_j)$ 不对 / t-test F-test 不对 / × efficient × BLUE

2. Test for serial correlation ① Test AR(1) with strict exo regressor (TS3)

1° $y_t = \beta_0 + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t$ 2° $\hat{u}_t = \rho \hat{u}_{t-1} + e_t$ u_t lag one

Symptotically reg $\hat{u}_t = \rho \hat{u}_{t-1} + error$ $H_0: \rho = 0$ → 无 serial correlation

② TS1 ~ TS6 Durbin-Watson Test (exact) $DW = \sum_{t=2}^{n}(\hat{u}_t - \hat{u}_{t-1})^2 / \sum_{t=1}^{n} \hat{u}_t^2 \approx 2(1 - \hat\rho)$

$H_0: \rho = 0$ → no serial corr $H_1: \rho > 0$

reject H_0 if $DW < d_L$ accept if $DW > d_U$, d_L d_U 之间 无结论.

③ 仅时候 General Breusch-Godfrey test for AR(g)

Symptotically reg $\hat{u}_t = \alpha_0 + \alpha_1 x_{t1} + \dots + \alpha_k x_{tk} + \rho_1 \hat{u}_{t-1} + \dots + \rho_g \hat{u}_{t-g} + error$

$H_0: \rho_1 = \dots = \rho_g = 0$ Joint test ⇒ no serial corr

3. 解决方法 ① Serial correlation-robust se (OLS se 会高估), 有效

HAC 方法 对 Heteroskedasticity & autocorrelation consistent (serial corr)

前提 大样本 ⇒ valid asymptotically newey w × lag(g) (含至 u_t & u_{t-g})

② FGLS 前提 u_t AR(1) error / 从 2 ① ② ③ 已得 $\hat\rho$

eg. $y_t = \beta_0 + \beta_1 x_t + u_t$ $y_t - \rho y_{t-1} = \beta_0(1-\rho) + \beta_1(x_t - \rho x_{t-1}) + u_t - \rho u_{t-1}$

$\rho y_{t-1} = \rho\beta_0 + \rho\beta_1 x_{t-1} + u_{t-1}$ $y_t^* = \beta_0(1-\rho) + \beta_1 x_t^* + e_t$

⇒ ∵ AR(1) $u_t = \rho u_{t-1} + e_t$ ⇔ $u_t - \rho u_{t-1} = e_t$ 无 serial correlati

且 e_t 一定满足 strict exogeneity (TS3) 才 consistent ∵ 变化 regressors 会不同

而 $cov(e_t, x_t, x_{t+1}) = cov(u_t - \rho u_{t-1}, x_t)$ 分开来 不相关 = 0 √ strict exo

transformed error/model 满足 G-M Asump BLUE   robust Se

③ 若 OLS robust & FGLS 差距大 → FGLS 的 TS.3 也不行 → 用 OLS

4. Pooled cross-section 每个 t 个体不同 (from random sampling of 总体)

有 (cross-section 中) ① Sample size ↑ $Var = \frac{\sigma^2}{SST_j(1 - R_j^2)}$ SST ↑ Var ↓ 精度

② 有 time effect: 加入 year dummy reg y $x_1 \dots x_g$ T_1 T_2 ... 有 benchmark

③ 加入 x_i × year dummy, 看 x_i 对 y 影响 Time 变化情况 缺: 太多 Dummy

5. Difference in difference (DID) for pooled cross-section data

Quasi-experiment Time 0 (处理前), 1 (后) / T (Treatment) & C (Control) 2个

$DID = (\bar{y}_{T,1} - \bar{y}_{C,1}) - (\bar{y}_{T,0} - \bar{y}_{C,0}) = (\bar{y}_{T,1} - \bar{y}_{T,0}) - (\bar{y}_{C,1} - \bar{y}_{C,0})$

剔除 随时间共同趋势 Treatment 的效果. $\hat\delta_1 = (\bar{y}_{T,1} - \bar{y}_{C,1}) - (\bar{y}_{T,0} - \bar{y}_{C,0})$

$price = \beta_0 + \delta_0 after + \beta_1 Treatment + \delta_1 after \cdot treatment + u$

⇒ policy evaluation 前提 not changed by other factor

§11 Panel Data

1. ① balanced or imbalanced (有 missing)

② same cross-sectional units followed over time   flexibility

③ Simple 2-period Panel Data model   → capture individual a_i 不随 t 变化

$y_{it} = \beta_0 + \delta_0 Time + \beta_1 x_{it1} + \dots + \beta_k x_{itk} + a_i + u_{it}$ heterogeneity

cross-section 的 每个个体 一样 ⇒ intercept $\beta_0 + a_i$ 每个个体有专用

2. Random effect → $cov(a_i, x_{itk}) = 0$ v_{it} 可看作 error term ⇒ OLS

Fixed effect → $cov(a_i, x_{itk}) \neq 0$ ⇒ endogeneity bias 不连续

3. Fixed effect 解决 ① First-differences: $\Delta y = \delta_0 + \beta_1 \Delta x_{i1} + \dots + \beta_k \Delta x_{ik} + \Delta u$

与时期 t 和 t−1 做 difference a_i 消失 ⇒ strict exo (Δu_i Δx_i) ⇒ unbias ⇒ OLS